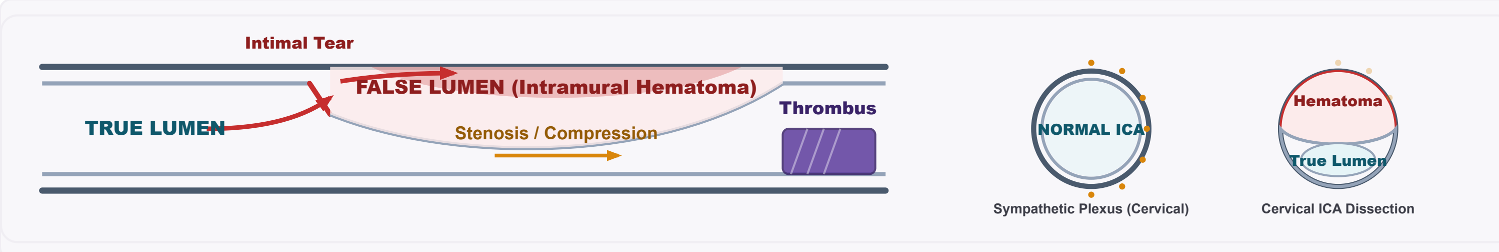
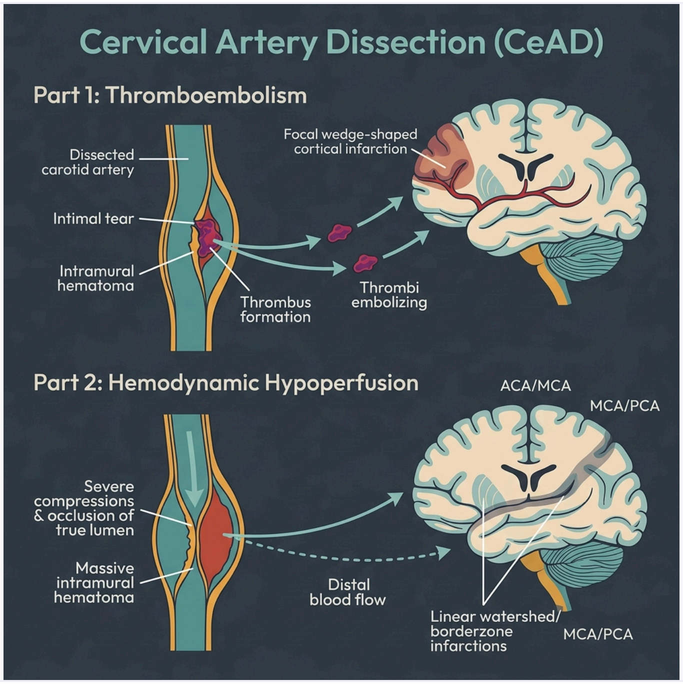
Cervical Artery Dissection

{/* Anatomy & Dissection SVG */}





1. Clinical Presentation & Pathophysiology

Ipsilateral Pain & Onset

• **Carotid (ICA)**: Frontotemporal/retro-orbital/facial pain (jaw angle).
• **Vertebral (VA)**: Severe occipital or posterior neck pain.
• **Onset**: Precedes stroke/TIA by hours to days (median 4 days).

Anhidrosis-Sparing Horner's

• **Signs**: Ptosis/miosis (28–58% of ICA) **without** anhidrosis.
• **Mechanism**: Sweat fibers follow ECA plexus; pupil/eyelid fibers follow ICA.

Neurological Deficits

• **CN Palsies**: CN IX–XII palsies (8–16%) from local ICA compression.
• **VA Territory**: Wallenberg syndrome, cerebellar ataxia, PICA/AICA strokes.

2. Diagnostic Workup

- **CTA Head/Neck**: Shows string sign, dissection flap, pseudoaneurysm, or occlusion.
- **MRI Neck (T1 Fat-Sat)**: Pathognomonic crescent sign (intramural hematoma).
- **DSA**: Reserve for diagnostic doubt or stenting.
- **Screening**: Assess for FMD/connective tissue disease if spontaneous/recurrent.

3. Medical Management: Extracranial vs. Intracranial Dissection

Extracranial Dissection

• **Antithrombotics**: ≥ 3 months (Class I).
• **Choice**: Equipoise. Monotherapy/DAPT vs. VKA/DOAC is individualized.
• **STOP-CAD**: In occlusions, consider anticoagulation Day 1–30, then switch to antiplatelet.
• **IV Thrombolysis**: Safe & indicated within 4.5 hours (Class I).

Intracranial & Pseudoaneurysms

• **SAH Risk**: Lack external elastic lamina & thin adventitia; high rupture risk.
• **Anticoagulation**: Contraindicated if SAH present. Screen with CT/LP first. Prefer single antiplatelet.
• **Pseudoaneurysm**: Conservative management with serial imaging.
• **Stenting**: Reserve for refractory ischemia or enlarging/symptomatic pseudoaneurysms.

4. Landmark Trial & Cohort Evidence

Study / Year	Population & Design	Interventions Compared	Key Outcomes & Clinical Nuance
CADISS 2015	N = 250. Extracranial CeAD. RCT.	Antiplatelet vs. Anticoagulant for 3 months.	• **Composite (Stroke/Death at 3m)**: 2.0% vs. 1.0% (p = 0.63). Established clinical equipoise.
TREAT-CAD 2021	N = 194 (PP = 173). Extracranial. RCT.	Aspirin 300mg daily vs. VKA for 3 months.	• **Composite (Stroke, bleed, death, or MRI at 14d)**: 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke: 8.0% vs. 0%.
Kaufmann IPD 2024	N = 444. Meta-analysis of CADISS + TREAT-CAD.	Antiplatelet vs. Anticoagulant.	• **Ischemic Stroke alone**: Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, p = 0.01). No difference in composite.
STOP-CAD 2024	N = 3,636. Multicenter cohort registry.	Antiplatelet vs. Anticoagulation.	• **Stroke vs Bleed**: Anticoagulation associated with lower stroke rate but higher bleed. Occlusions benefited most; day 30 transition.

CADISS: *Lancet Neurol*. 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD:** *Lancet Neurol*. 2021;20(5):341-350. PMID: 33765420
Kaufmann IPD: *JAMA Neurol*. 2024;81(6):630-637. PMID: 38739383 | **STOP-CAD:** *Stroke*. 2024;55(4):908-918. PMID: 38334460 | **AHA/ASA:** *Stroke*. 2021;52:e364-e467. PMID: 34024117